COMMERCIAL ITEM DESCRIPTION

MAGNETIC PARTICLE PROCESS CONTROL KIT

The General Services Administration has authorized the use of this commercial item description.

Scope. The magnetic particle process control kit contains the components necessary for testing new and in-use magnetic particle inspection materials to determine the degree of contamination/degradation of these materials. The kit will also be employed to test for degradation in the magnetic particle inspection equipment and magnetic particle inspection process. The results of the process control inspection verify the performance characteristics of the magnetic particle inspection materials and equipment to determine their suitability for use in a magnetic particle inspection. In addition, the inspection validates the integrity of the inspection process used by the nondestructive inspection (NDI) technician.

APPLICABLE DOCUMENTS

MIL-M-6867	Magnetic Inspection Units.
MIL-STD-1949	Inspection, Magnetic Particle.

American Society for Testing and Materials (ASTM)

ASTM D-96-73	Water and Sediment in Crude Oils.

(Application for copies shall be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Salient characteristics: The magnetic particle process control kit shall be comprised of the following components

1. Steel ring. A steel ring with known discontinuities fabricated from AISI 01 tool steel cut from annealed round stock. The steel ring shall have a Rockwell B hardness between 90 and 95. The steel ring shall have dimensions as shown in MIL-STD-1949, Figure 1. Tolerances for the ring's overall diameter, the ring's width, the diameter of the ring's center hole, and the distance between adjacent discontinuities shall be +0.762 mm. Tolerances for all other dimensions shall be ±.127 mm. The finish of the peripheral surface (surface where fluorescent indications are viewed) shall be 3.175 micron RMS or better. The contractor shall provide written certification that the steel ring was fabricated from the materials described above. The following procedures shall be used for all steel ring to correct improper annealing during manufacturing process.

a. Heat in furnace to 1450 degrees F (790 degrees C).

b. Hood at 1450 degrees F (790 degrees C) for 3 hours.

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c. Furnace cool at a rate of not more than 40 degrees F (22 degree C) per hour.

d. Upon reaching 1000 degrees F (54 degrees C), remove from furnace and air cool.

e. After the ketos ring has air cooled, remove any scale or discoloration by grit blasting with 100 mesh aluminum oxide, or 80 grit glass beads.

f. Protect the ketos ring by applying a rust and corrosion preventative and put the ring into a plastic bag.

2. Centrifuge tube. A 100 ml, pear-shaped graduated centrifuge tube as shown in ASTM D-96, Figure 6.

3. Centrifuge tube stand. A rigid stand used to support the centrifuge tube described in 2. above. The stand shall securely support the centrifuge tube in the vertical position while permitting full view of the tube. The stand shall be constructed of a nonmagnetic, oil-resistant material. Additionally, the stand shall make provisions for supporting a horseshoe magnet (see 4. below) so that the magnet's poles can make contact with the centrifuge tube's outer surface at the point where the tube attains its largest diameter.

4. Magnet. A horseshoe magnet with dimensions that enable both poles to fit flush against the side of the centrifuge tube when the centrifuge tube and magnet are supported by the centrifuge tube stand (see 3.)

5. Test meter kit, digital. A test kit capable of measuring the output current of type II wet, stationary, magnetic particle inspection systems (see MIL-M-6867) in the contact mode, both alternating current (AC) and full-wave direct current (FWDC). The test kit shall include a digital meter that provides a direct readout of the output current in amperes over a minimum range of 0 to 10,000 amperes, both AC root mean square (RMS) and FWDC The meter shall incorporate a hold and display feature that retains the current reading until the next magnetizing shot. The digital meter shall have an accuracy of ±1 percent. The test kit shall be capable of calibration traceable to the National Bureau of Standard (NBS). The contractor shall provide calibration procedures to the Government

6. Photometer. A portable, hand-held photometer used for measuring the intensity of visible light inside the magnetic particle inspection booth The photometer shall measure visible light intensities in a minimum range of 0 to 5 foot candles or 0 to 54 lux. The photometer shall be scaled in foot candles, or lux. Resolution of the meter shall be 0.5 foot candles, or 5 lux or better. The meter shall be capable of calibration within +2% of an NBS traceable standard. The contractor shall provide calibration procedures to the Government The technical order of the photometer shall have a table showing the conversion between foot candle and lux.

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